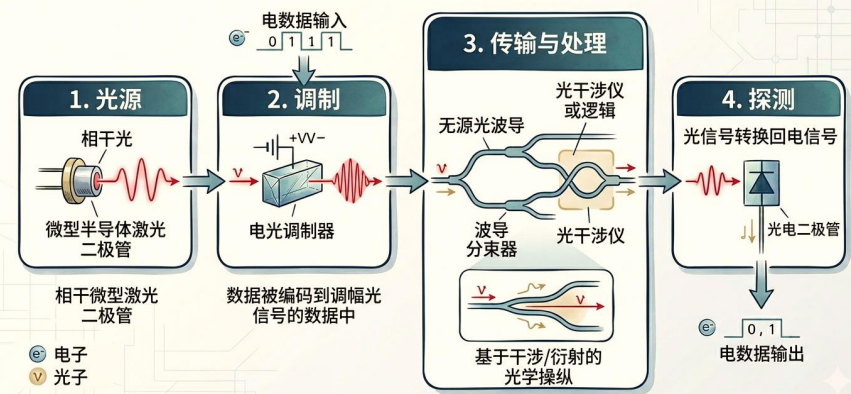


[光芯片系统定位]

可编程光子集成芯片（PIC）是多器件集成、强耦合、易受环境影响的复杂动态控制系统，是光通信、光计算、信号处理的核心硬件。

光芯片系统基本原理



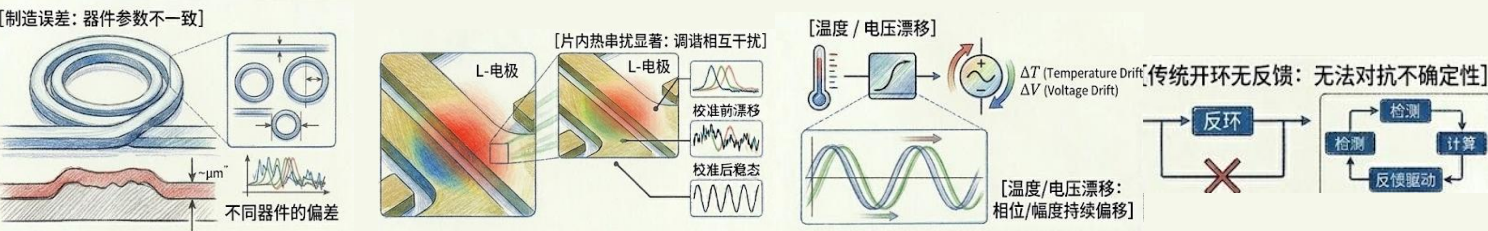
[控制论核心问题]

系统天然存在不确定性扰动与熵增趋势，自发走向失序与失稳，必须依靠闭环反馈实现秩序构建与稳态维持。

[解决核心问题]

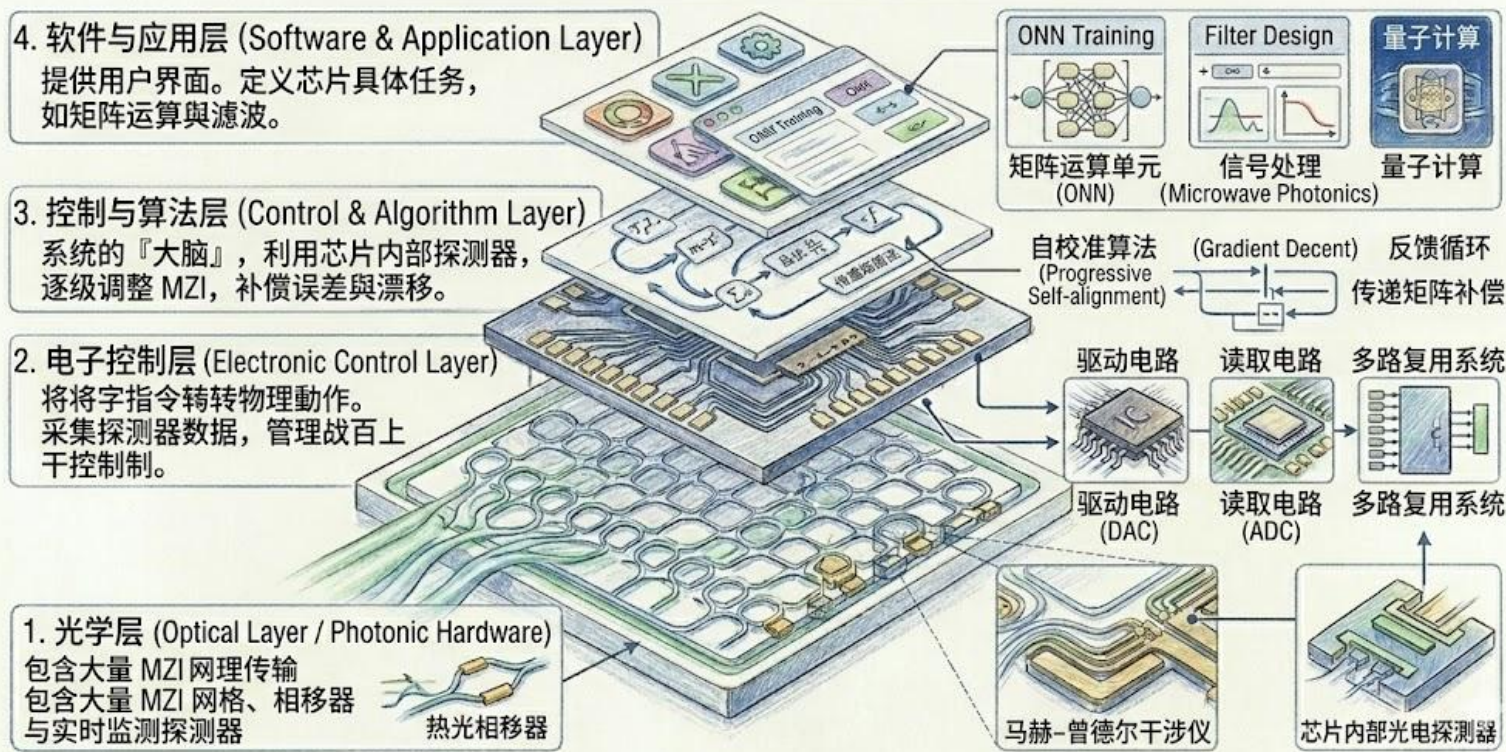
通过闭环控制系统，让光芯片在众多干扰下实现自校准

[工程痛点]



制造误差导致器件参数不一致
片内热串扰显著，调谐相互干扰
温度、电压漂移引发相位、幅度持续偏移
传统开环控制无反馈、无修正，无法对抗不确定性

完整光芯片自校准系统分层结构图



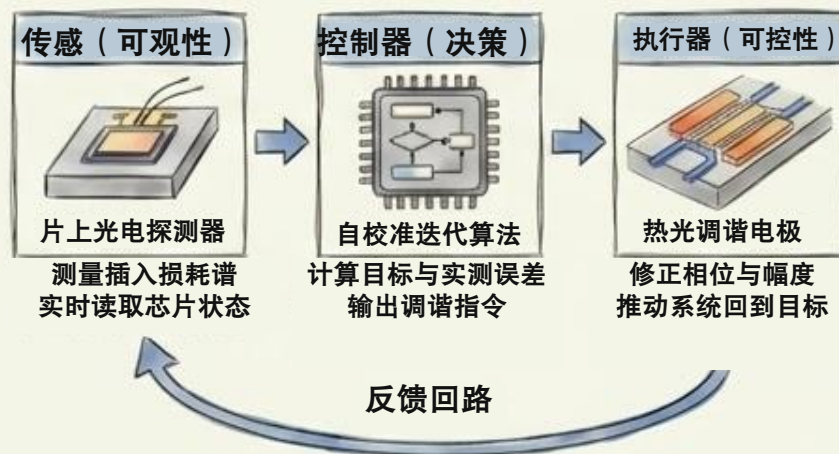
自校准闭环控制：感知 - 反馈 - 调节 - 收敛

“方法如何落地”：把控制论里的可观、可控、反馈，转成光芯片可执行的工程流程。

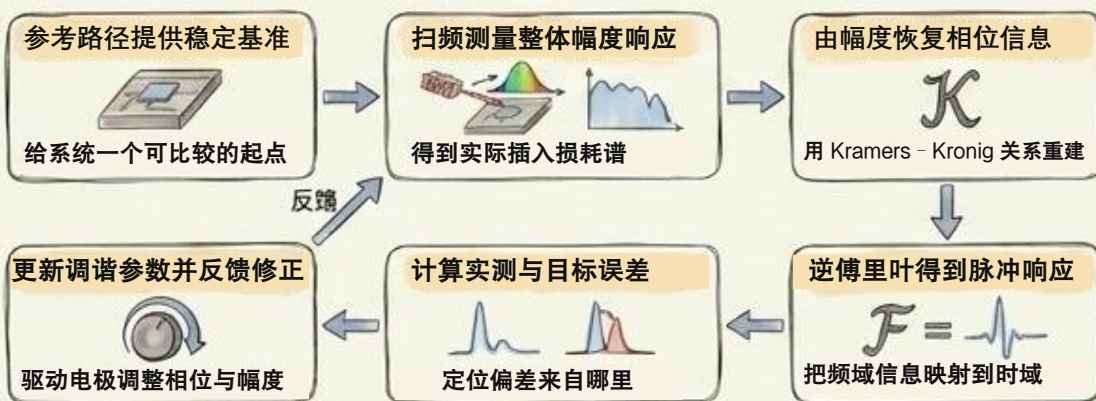
[控制目标]

- 在无先验模型、无人工干预条件下，自动补偿热串扰与参数漂移

- ✓ 光功率恒定
- ✓ 波长精准锁定
- ✓ 相位同步匹配
- ✓ 工作温度稳态
- ✓ 热串扰自动补偿
- ✓ 目标稳态

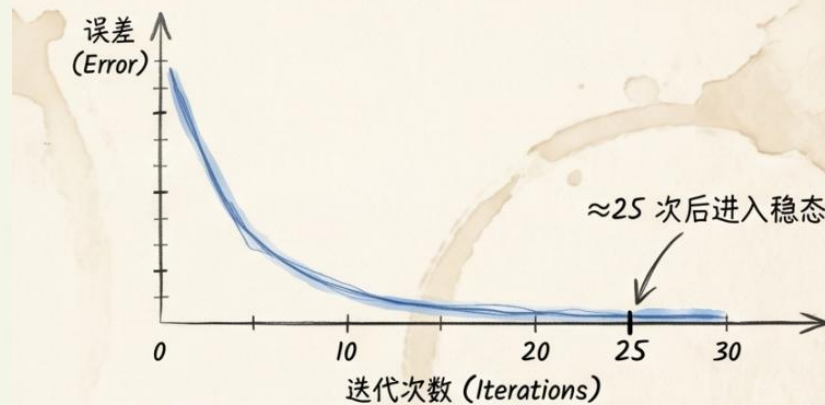


自校准完整流程



论文核心链路：先测幅度，再恢复相位，随后根据误差更新调谐参数并反复迭代。

[关键结果] / [收敛结果]



模型无关

全局闭环

自适应抗扰

快速收敛

误差随迭代次数快速下降，体现模型无关、全局闭环和自适应抗扰的工程优势。

[光芯片里的控制哲学]



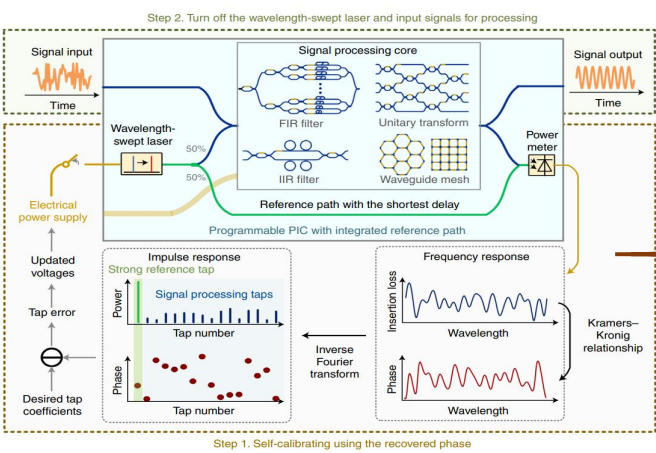
具体体现



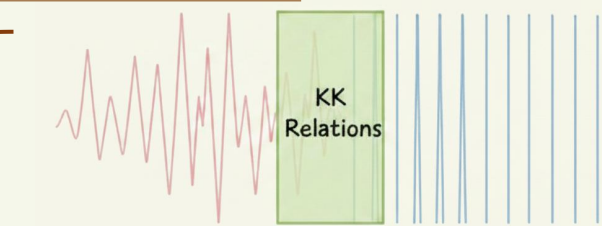
整体框图

秩序的演化

- 控制论解决的不仅是光芯片的制造误差，更是复杂集成系统自发的“**熵增趋势**”。
- 自校准过程**是人类意志通过数学逻辑，对物理世界无序状态的强行介入。我们不追求物理实体的绝对完美，而是在信息的循环流动中构建逻辑的稳态。

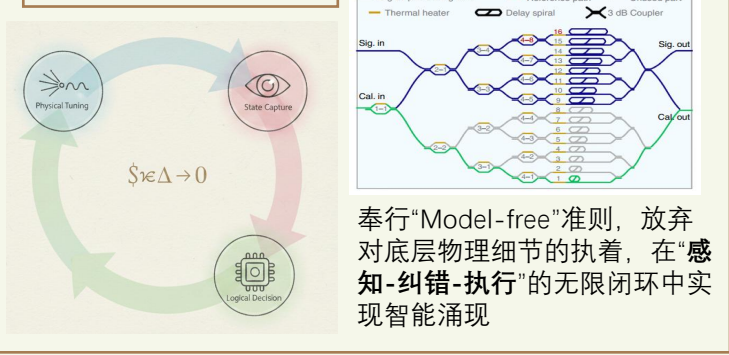


确定性对立统一



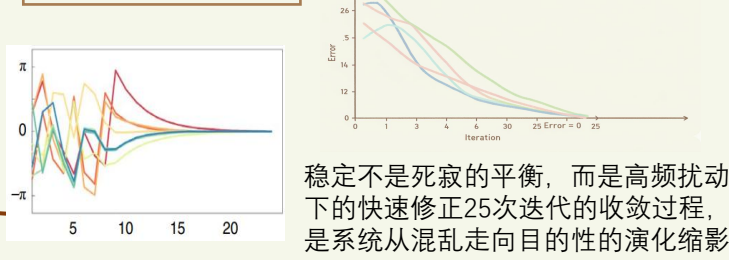
利用 **Kramers-Kronig** 关系的因果逻辑，从残缺的强度信息中还原出完整的相位秩序

反馈本质



奉行“Model-free”准则，放弃对底层物理细节的执着，在“**感知-纠错-执行**”的无限闭环中实现智能涌现

动态稳态



稳定不是死寂的平衡，而是高频扰动下的快速修正25次迭代的收敛过程，是系统从混乱走向目的性的演化缩影

系统观与方法论总结

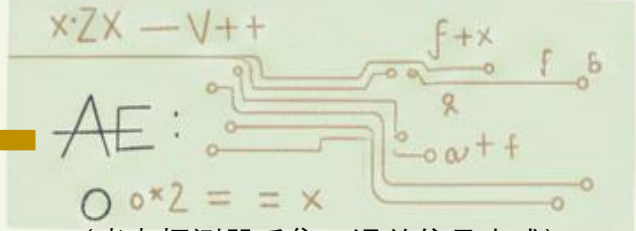
- 整体论**：解决强耦合矛盾，需放弃局部执着，追求系统全局最优。
- 反馈论**：与其追求“全知全能”的模型预判，不如深耕“知错就改”的响应机制。
- 稳态观**：真正的稳定不是静止，而是在持续扰动中不断修正的动态平衡。

(系统论、控制论思维导向)



决策哲学层

(25次迭代算法、KK相位恢复)

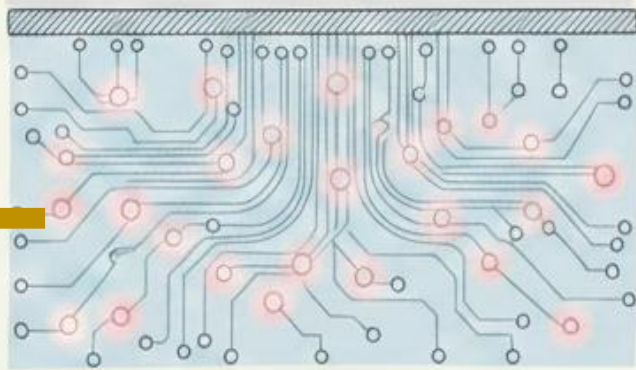


算法策略层

(光电探测器采集、误差信号生成)



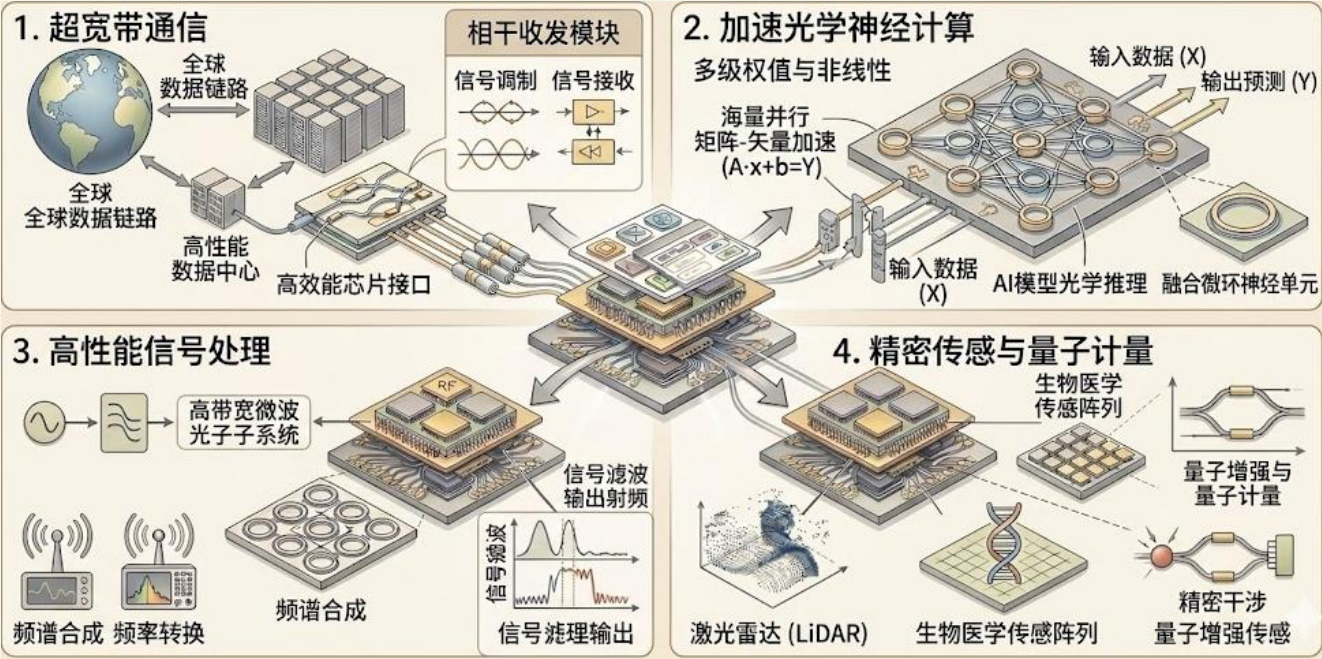
信息反馈层



物理执行层

(热光相移器、MZI网格物理调整)

[光芯片应用场景]



[文献支持]

Self-calibrating programmable photonic integrated circuits
Xingyuan Xu^{1,2,3}, Guanghui Ren³, Tim Felleppa¹, Xumeng Liu¹, Andreas Boes^{3,4}, Anran Mitchell³ and Arthur J. Lowery¹
Programable photonic integrated circuits (PICs) are dense assemblies of tunable elements that provide flexible reconfigurability to enable different functions to be selected; however, due to manufacturing variations and thermal gradients that affect the optical phases of the elements, it is difficult to guarantee a stable correspondence between the electrical commands to the chip, and the function that it provides. Here we demonstrate a self-calibrating programmable PIC with full control over its complex impulse response, in the presence of thermal cross-talk between phase-tuning elements. Self-calibration is achieved by: (1) incorporating an optical reference path into the PIC; (2) using the Kramers-Kronig relationship to recover the phase response from amplitude measurements; and (3) applying a fast-converging self-calibration algorithm. We demonstrate full control over signal processing functions with complex impulse responses using only 25 training iterations. This approach offers stable and accurate control of large-scale PICs, for demanding applications such as communications network reconfiguration, neuromorphic hardware accelerators and quantum computers.

[1] Xu, X., Ren, G., Felleppa, T. et al. Self-calibrating programmable photonic integrated circuits. Nat. Photonics 16, 595–602 (2022).
<https://doi.org/10.1038/s41566-022-01020-z>

汇报人: [REDACTED]

[结论]

课程思想总结

- 控制：在不确定世界中构建秩序、对抗熵增
- 反馈：系统实现稳定与智能的核心
- 稳态：动态平衡，而非绝对静止
- 控制既是工程方法，也是理解复杂系统的哲学思维

控制方法通用性

适用于所有复杂动态系统：精密控制、自动化、通信系统、集成电路、生物稳态系统。

文献核心验证

- 光芯片实用化的关键：闭环反馈 + 自校准
- 反馈控制可有效对抗不确定性、热串扰、参数漂移
- 自校准是典型模型无关自适应控制